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Mispricing in CNX Nifty Futures: An Empirical Investigation

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Abstract

Derivatives markets have witnessed considerable growth since their inception at Indian stock exchanges. In the present study, we aim to assess the pricing efficiency of the CNX Nifty index futures. We compare the theoretical cost of carry price with the corresponding observed prices for the period January 2003 to December 2012. In an efficient market, these two are expected not to differ, since such a deviation could lead to riskless arbitrage opportunities. Non-parametric tests are employed to test the statistical significance of our results. We report that the CNX Nifty futures are frequently underpriced, possibly due to short-sales restrictions. There are instances of overpricing, too, but the frequency and magnitude of such scenarios are fewer. We find significant differences in the behaviour of mispricing between the instances of negative/positive basis. Similarly, there appears to be a significant difference between mispricing series with changing days to expiry. These findings can have important implications, which are presented in the conclusion along with an outline of future prospects for research.

Keywords

Mispricing, index futures, market efficiency, cost-of-carry, basis, time-to-maturity

Introduction

This article is an attempt to investigate the pricing efficacy of the S&P Nifty index futures. CNX Nifty is the flagship index on the National Stock Exchange of India Ltd. (NSE). The index tracks the behaviour of a portfolio of 50 blue-chip companies, covering 22 sectors of the Indian economy (www.nseindia.com).

Derivatives trading in India commenced in June 2000 with trading in stock index iutures. Since then, there has been a huge growth in terms of number of contracts traded and turnover (refer to Table 1). The surge in volumes necessitates that investor interest is taken care of without any compromise in growth prospects.

At the same time, it is important that the futures market perform the function of price discovery and risk management with utmost efficiency. The hedging effectiveness of futures contracts can be compromised if it the market lacks pricing efficiency. We operationalise the cost of carry model and compare the prices so arrived at, with the observed prices of the corresponding futures contracts.

Table 1. Business Growth in Index Futures at National Stock Exchange (NSE)

Years	No. of Contracts	Turnover (₹ crore)	YOY Growth in Turnover
2000–2001	90,580	2,365	
2001–2002	1,025,588	21,483	808%
2002–2003	2,126,763	43,952	105%
2003–2004	17,191,668	554,446	1161%
2004–2005	21,635,449	772,147	39%
2005–2006	58,537,886	1,513,755	96%
2006–2007	81,487,424	2,539,574	68%
2007–2008	156,598,579	3,820,667	50%
2008–2009	210,428,103	3,570,111	−7%
2009–2010	178,306,889	3,934,389	10%
2010–2011	165,023,653	4,356,755	11%
2011–2012	146,188,740	3,577,998	−18%
2012–2013	96,100,385	2,527,131	−29%

Source: www.nseindia.com

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Any deviation between the theoretical price and the observed price would signify presence of arbitrage opportunities, and thereby amount to mispricing. Such an opportunity to make risk-free profits is counter-intuitive to the notions of market efficiency.

There have been a number of studies with respect to futures mispricing, especially in the US and European markets. However, studies in the Indian context are scant and cover shorter study periods on account of the newness of the market. The present study is an attempt to bridge this gap by extending the previous studies in terms of time period, and thereby commenting on the trend in the market over the study period.

This article comprises seven sections including this section, the introduction. The previous researches conducted on the topic are presented in the second section. The next section lays down the theoretical framework for the study. The fourth section describes the data and research methodology. The empirical findings are presented in the fifth section, followed by statistical testing. Finally, the seventh section concludes the article, also outlining the limitations of the study and the scope for further research.

Literature Review

The pricing efficiency of futures markets has been examined in several markets. A review of past literature is presented below according to the nature of their findings.

Evidence of Mispricing

MacKinlay and Ramaswamy (1988) utilised time-stamped data for the S&P 500 in the US market to examine spot-future arbitrage. The study reported that variability in futures prices exceeded variability in spot prices, which remained even after considering non-synchronous trading.

In the German market, Bühler and Kempf (1994) observed significant mispricing signals. It was observed that the number and persistence of arbitrage opportunities differs considerably for futures nearest to deliver as compared to futures which are not nearest to deliver.

Brailsford and Hodgson (1997) constructed and analysed a mispricing series using contemporaneous observations from the Australian index futures market over a period of five years. The results indicated mean underpricing, consistent with the more expensive transaction cost boundary which involved short-selling of stocks.

Using both intra-day and inter-day data, Garrett and Taylor (2001) observed predictability in mispricing changes for FTSE 100 at LIFFE. The study showed that this predictability was neither due to microstructure effects nor due to arbitrage activity.

In the Indian market, Vipul (2005) explored the variation of the pricing inefficiency of the stock and index futures with time for the Nifty index futures and stock futures from January 2002 to November 2004. The Nifty index futures were persistently underpriced due to restrictions on short selling and high volume stock futures were generally overpriced. The findings were in line with international experience for index futures, although stock futures exhibited different behaviour.

Misra, Kanan and Misra (2006) corroborated with the above and observed violation of spot future parity relationship for NSE's CNX Nifty index futures for a two-year period. They also reported that arbitrage profits are more for far-month future contracts than near-month future contracts, for undervalued future markets than overvalued future markets, for highly liquid futures than for less liquid futures.

In another study by Vipul (2008), the relationship between mispricing and various other variables was explored. In line with the earlier findings, numerous incidents of mispricing were observed. The mispricing persisted for one day. The article also reported that a change in futures mispricing did not have a predictable effect on the financial activities for stock futures.

Gupta and Singh (2009), in their study spanning four years, found that a strong and stable relationship existed between Indian equity futures and cash markets in the long run but, in the short run significant deviations were observed. Empirical findings suggested that more arbitrage opportunities were available near the expiration date.

In Turkey, MacMillan and Ülkü (2009) reported persistent and typically negative mispricing. This was attributed to short-selling effects, behavioural effects and insufficient arbitrage. Fassas (2010) investigated the FTSE/ATHEX 20 Index Futures mispricing in the Greek market. Significant mispricing was found, and the author opined that the arbitrage transactions are not as strong and effective in the Greek market as they are in other mature markets (S&P 500 and FTSE 100 markets).

No Evidence of Mispricing/Small Magnitude of Mispricing/Mispricing Reduces after Incorporating Market Imperfections

Chung (1991) investigated the Major Markets Index (MMI) at the Chicago Board of Trade (CBOT) and contradicted previous studies by reporting a substantially smaller frequency and magnitude of mispricing.

Neal (1996) tested arbitrage models using the S&P 500 and concluded that short-sale restrictions were unlikely to affect the futures mispricing, and the magnitude of mispricing was small.

In the context of the Hong Kong Market, Fung and Draper (1999) conducted a study over three distinct regulatory regimes relating to the short-selling of stocks. They reported significant mispricing. However, relaxing the constraints on short-selling reduced the extent of futures mispricing. This was an important finding especially in context of such markets where short-sales restrictions exist.

Cummings and Frino (2011) studied the Australian stock index futures and indicated that mean pricing error was close to zero confirming that a tax-adjusted market valuation model produces a relatively unbiased estimates for the future prices. In China, Wang (2011) examined the cost of carry model for the GX FTSE Xinhua China A50 and HKEx H-share index futures markets. The model could not predict accurately the price of these futures on account of high volatility in the cash markets. After incorporating volatility in the model, the predictive accuracy improved. On the contrary, Tharavanij (2012), while examining the Thai stock index futures contract, found that cost-of-carry model explained futures pricing very well.

From the discussion above it is evident that futures pricing has been investigated globally. There has been some conflict in the findings regarding evidence of mispricing; however, most of the studies point towards significant mispricing in futures market, using a cost of carry model. Studies pertaining to the Indian market are fewer and span a shorter time period perhaps on account of the newness of the segment. Our article attempts to bridge this gap by examining the behaviour of mispricing for a 10-year period and thereby commenting on the trend for CNX Nifty index futures.

Pricing of Index Futures Contract

We lay down the theoretical framework in the present section within which we have modelled our study. Since only index futures are studied, the ensuing discussion is restricted to the pricing of index futures only. According to the cost-of-carry model, the fair price of a futures contract should be equal to the underlying index price adjusted for the cost of carrying it for the remaining life of the contract. It can be denoted with the following equation:

$$F_0 = S_0 e^{(r-q)T} \quad (1)$$

where,
F_0 = Futures price today
S_0 = Spot price of the index today
r = Risk free interest rate
q = Annualised dividend yield on the index, continuously compounded
T = Time to expiry in years

The cost-of-carry includes the interest cost and the dividend paid on the underlying. For the purpose of our study, we record mispricing when there is a deviation between the fair price determined by implementing the cost-of-carry model and the observed market price. A positive (negative) mispricing connotes that the index future is overvalued (undervalued), that is, the observed price is more than (less than) the fair price.

Whenever there is a departure of the observed price from its fair price, arbitrage opportunities arise. In the case of positive mispricing, an arbitrageur makes a risk-free profit by buying the spot index portfolio and opening a short position in the futures market (long arbitrage). On the other hand, if the futures are underpriced, the arbitrageur can shorten the index, and buy the futures (short arbitrage). At expiry, the riskless profit is equal to the difference between the observed price and the fair price. The cash flows in both the situations are presented below. In addition to the above-mentioned notations, we use F_I to denote the observed futures price in present time period. For simplification, we have left out the dividend yield from the example.

A cautionary note needs to be presented here. After considering transaction costs, these profits may not exist at all. It is also evident from the above discussion that in case of negative mispricing, it is difficult to put the arbitrage strategy into place since it requires short selling of the index, which is restricted in many markets, including India.

Table 2a. Arbitrage Strategy (Long)

When Observed Price (F_I) > Fair Price (F_0)	Now	At Expiry
Buy index	$-S_0$	$+F_I$
Sell futures	$+F_I$	$-F_I$
Borrow cash	$+S_0$	$-S_0e^{rT}$
Payoff	F_I	$-S_0e^{rT}$
Total payoff	$F_I-S_0e^{rT}$	

Source: Authors.

Table 2b. Arbitrage Strategy (Short)

When Observed Price (F_I) < Fair Price (F_0)	Now	At Expiry
Sell index	$+S_0$	$-F_I$
Buy futures	$-F_I$	$+F_I$
Loan cash	$-S_0$	S_0e^{rT}
Payoff	$-F_I$	S_0e^{rT}
Total payoff	$S_0e^{rT}-F_I$	

Source: Authors.

Research Design

Objectives of the Study

The main objectives of the study are as follows:

1. To assess the signals of mispricing in CNX Nifty over the sample period January 2003 to December 2012.
2. To discern the trend for the mispricing (if any) over the sample period.
3. To analyse the mispricing (if any) when the basis is negative (Spot Price < Observed Futures Price) and when the basis is positive (Spot Price > Observed Futures Price).
4. To study the behaviour of mispricing (if any) across different levels of maturities.

Data and Methodology

Our study conducts ex-post analysis, that is, a study of mispricing on the transaction date only. Wherever deemed necessary, the study period has been divided into two subperiods: January 2003 to December 2007 and January 2008 to December 2012. The magnitude of mispricing is studied individually, for positive signals and negative signals, and also in terms of overall mispricing.

Along the lines of Vipul (2008), only near-term futures contracts are taken into account since the next-month and far-month contracts lack liquidity. The data for the expiry day has been filtered out from the sample as there has been evidence of increased volatility due to traders winding up their positions on the expiry day (Debashish & Puri, 2010; Vipul, 2005).

The data for the present study consists of three sets: (*a*) data related to futures contracts like deal date, expiry date, daily closing futures price and number of contracts traded; (*b*) data pertaining to the daily spot index, like the price of the underlying and dividend yield; and (*c*) the risk-free interest rate, for which we have used 30-day Mumbai Inter-Bank Offer Rate (MIBOR). Since banking holidays do not match stock exchange holidays, MIBOR is unavailable for certain transaction dates. For such days, previous days' MIBOR has been considered. All the above-mentioned datasets have been taken from the official website of NSE (www.nseindia.com). The dividend yield and risk-free interest rates have been continuously compounded.

The total number of observations for the sample period was 2498. After applying the expiry day filter, the observations were reduced to 2382 daily data points. A year-wise breakup can be observed from Table 3. We do not take transaction cost and bid–ask spread into account in our study. From the standpoint of interpretation, however, an attempt has been made to interpret the results in light of the presence of both these factors.

Results and Discussion

Frequency and Magnitude of Mispricing

Equation (1), specified in the third section, has been operationalised to calculate the extent of mispricing. A comprehensive set of results has been presented in Table 3. The total number of observations for the sample

Table 3. Mispricing (₹) in CNX Nifty for the Period January 2003 to December 2012

Year & Mispricing	No.	Mean	Standard Deviation	Min	Max	q1	q2	q3
2003	**241**	**−1.0145**	**4.8710**	**−14.7058**	**10.4874**	**−3.8712**	**−0.5720**	**2.3268**
Negative	136	−4.2365	3.6751	−14.7058	−0.0190	−6.3159	0.0000	−1.3036
Positive	105	3.1589	2.4618	0.0942	10.4874	1.3409	2.7749	4.2947
2004	**245**	**−5.6205**	**8.8215**	**−52.1272**	**16.6585**	**−10.2033**	**−4.3842**	**0.0816**
Negative	181	−8.8093	7.8215	−52.1272	−0.1145	−11.8916	−6.3030	−3.3742
Positive	64	3.3976	3.8439	0.0146	16.6585	1.1299	1.9859	4.0436
2005	**240**	**−9.7651**	**9.3989**	**−36.2311**	**9.2452**	**−16.4305**	**−9.0385**	**−2.4885**
Negative	199	−12.3420	8.1598	−36.2311	−0.0236	−17.4368	−11.5696	−5.8178
Positive	41	2.7427	2.2032	0.1109	9.2452	0.8053	2.2320	3.4821
2006	**238**	**−15.4230**	**15.7293**	**−82.8656**	**23.2186**	**−24.4873**	**−14.1261**	**−3.9217**
Negative	197	−19.5051	14.0786	−82.8656	−0.5315	−25.8884	−17.3751	−9.4098
Positive	41	4.1908	4.2792	0.0314	23.2186	1.5750	2.8746	6.1528

Year & Mispricing	No.	Mean	Standard Deviation	Min	Max	q1	q2	q3
2007	**236**	**−17.9562**	**19.7983**	**−77.7546**	**30.6083**	**−31.7386**	**−16.9297**	**−5.2386**
Negative	192	−24.2994	15.9711	−77.7546	−0.0011	−33.6823	−21.2869	−12.0448
Positive	44	9.7233	6.7477	0.2449	30.6083	4.9776	8.4247	12.9938
2008	**234**	**−14.5441**	**21.0947**	**−94.6190**	**28.8170**	**−23.2766**	**−10.8857**	**0.3785**
Negative	174	−21.9267	19.3566	−94.6190	−0.3893	−28.5638	−16.6823	−7.8517
Positive	60	6.8655	5.6398	0.2413	28.8170	2.1554	5.6418	10.4443
2009	**236**	**−5.9118**	**10.4904**	**−42.3660**	**21.4431**	**−11.9399**	**−4.6423**	**0.6999**
Negative	166	−10.5469	8.7235	−42.3660	−0.1105	−15.3126	−8.3518	−4.0056
Positive	70	5.0801	4.3093	0.0273	21.4431	1.4390	4.1497	7.9209
2010	**239**	**−4.1705**	**12.2214**	**−43.0462**	**36.4987**	**−11.9133**	**−4.3504**	**3.3027**
Negative	159	−10.7045	8.1828	−43.0462	−0.1175	−14.6381	−9.1412	−4.4042
Positive	80	8.8157	7.7046	0.2383	36.4987	3.3988	6.4624	11.1650
2011	**235**	**−7.5941**	**11.4247**	**−46.8492**	**18.5603**	**−14.4272**	**−6.0010**	**−0.0730**
Negative	178	−11.8860	9.5242	−46.8492	−0.0220	−17.2652	−9.8757	−4.6108
Positive	57	5.8088	4.1582	0.0216	18.5603	2.0999	5.8652	8.8451
2012	**238**	**−0.2281**	**12.5936**	**−42.9656**	**30.4210**	**−6.4182**	**1.3293**	**7.9555**
Negative	110	−10.5354	10.0334	−42.9656	−0.0169	−15.2720	−7.1755	−2.9099
Positive	128	8.6298	6.2021	0.1135	30.4210	3.5554	7.3784	11.9453
2003–2007	**1200**	**−9.8946**	**14.2257**	**−82.8656**	**30.6083**	**−17.0200**	**−6.5217**	**−0.1095**
Negative	905	−14.5135	13.2234	−82.8656	−0.0011	−20.7691	−11.2334	−4.4190
Positive	295	4.2754	4.5282	0.0146	30.6083	1.3348	2.8748	5.4976
2008–2012	**1182**	**−6.4587**	**14.8199**	**−94.6190**	**36.4987**	**−2.7165**	**4.6025**	**13.6553**
Negative	787	−13.3960	12.9322	−94.6190	−0.0169	−9.9771	−6.2597	−2.7071
Positive	395	7.3633	6.0858	0.0216	36.4987	4.6134	10.2185	17.7511
2003–2012	**2382**	**−8.1896**	**14.6219**	**−94.6190**	**36.4987**	**−15.1898**	**−5.4547**	**1.0560**
Negative	1692	−13.9937	13.0968	−94.6190	−0.0011	−19.3732	−10.5127	0.0000
Positive	690	6.0431	5.6803	0.0146	36.4987	1.9236	4.3538	8.4829

Source: Authors.

period is 2382. The futures market seems to be underpriced compared to the fair price since the overall mispricing is ₹ −8.1896. The positive mispricing signals were 690 (28.97 per cent) whereas the negative signals were 1,692 (71.03 per cent). A higher proportion of negative mispricing indicates that the profits were not arbitraged away, which could be due to short-selling restrictions at Indian bourses (refer to the third section). Further, the magnitude of mispricing is also much higher (₹ −13.9937) for negative signs than for positive signs (₹ 6.0431). This denotes that there could be significant exploitable mispricing signals when there is under-pricing in the futures market.

These findings are in consonance with previous studies. In the German market (Bühler & Kempf, 1994), the DAX futures are underpriced for all sub-periods. Brailsford and Hodgson (1997) reported overall mean negative mispricing error in the Australian market with a 52.7 per cent of negative signals for total observations. In the newer Turkish market, too, McMillan and Ülkü (2009) observed persistent under-pricing of index futures.

In the Indian context, Vipul (2005) established underpricing of Nifty futures to the extent of 0.72 per cent for NSE members and 1.10 per cent for FIs after accounting for the transaction costs. Misra et al. (2006) also evidenced under-pricing in Nifty index futures.

Studying the sub-periods, there were 905 (75.42 per cent) observations of negative mispricing for the period 2003–2007, whereas the positive mispricing signals

were for 295 (24.58 per cent) instances. For the next five-year period, the frequency of under-pricing was 787 (66.58 per cent) and overpricing was 395 times (33.42 per cent). Clearly, for both the sub-periods, the CNX Nifty futures seem to be more frequently underpriced; however, for 2008–2012, the proportion of negative mispricing has reduced. This signifies that the short-selling restriction cannot be the only explanation for inconsistencies in pricing.

Looking at the two sub-periods, the magnitude of positive mispricing has also increased in the period 2003–2007 from ₹ 4.2754 to ₹ 7.3633 in 2008–2012. However, both would be un-exploitable after taking transaction cost into consideration. In case of negative mispricing signals, the magnitude has decreased in terms of absolute rupee terms—from ₹ −14.5135 to ₹ −13.3960. These can lead to profitable arbitrage opportunities if the liquidity is high in the market.

Analysing the pattern year-wise, Table 3 reveals interesting results. For the first five years of the sample period, the overall magnitude of mispricing rose consistently.

Considering under-pricing first, the frequency of the same increased from about 56 per cent of the total observations in 2003 to about 81 per cent in 2007. However, the magnitude of negative mispricing witnessed an upward trend, from ₹ −4.2365 in 2003 to an alarming ₹ −24.2994 in 2007, signifying considerable arbitrage opportunities, which could not be exploited due to short-sales restrictions. Looking at the positive mispricing, perceptibly, the proportion has decreased over the first sub-period; but the magnitude of mispricing has shown an increase from ₹ 3.1589 to ₹ 9.7233 (refer to Figure 1). With the exception of the year 2005, there was an increase in all the other years.

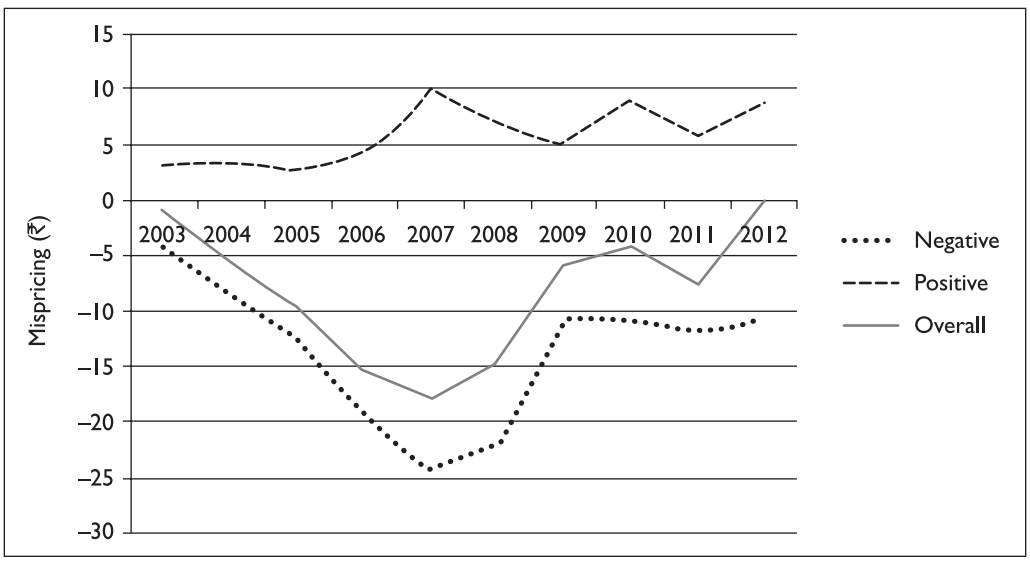
On investigating the second sub-period, 2008–2012, we observe that the amount of mean overall mispricing showed a consistent decline with the exception of the year 2011. The mean mispricing went down from ₹ −14.5441 in 2008 to ₹−0.2281 in 2012. This denotes that the observed prices of Nifty futures are much in line with their fair price in the recent years. The mean negative mispricing was at ₹ −21.9267 in 2008, which nearly halved to ₹ −10.5354 in 2012. This could offer some profitable arbitrage opportunities even after considering transaction costs. The positive mispricing did not show a steady pattern in the second sub-period (refer to Figure 1).

The year 2012 presents itself as an anomaly year when the proportion of positive mispricing was more than negative mispricing. This upholds our observation earlier that short-sales restrictions do not explain all mispricing in CNX Nifty futures. This can be a cause of concern, because the hedging effectiveness of the futures market gets undermined when it lacks pricing efficiency.

Mispricing with Respect to Days to Expiry (DTE)

We have charted mispricing signals for various days left to expiry. The results, presented in Table 4, reveal that mean mispricing reduces when there is a fewer number of DTE. When DTE is 21 days or more, the magnitude

Figure 1. Mispricing (₹) for the period 2003–2012



Source: Authors.

Table 4. Mispricing with Respect to Days to Expiry for the Period 2003–2012

Days to Expiry	Observations	Mean	Standard Deviation	Min	Max
1 to 7	**573**	**−0.4645**	**9.6470**	**−61.6495**	**30.6083**
Negative	270	−7.7476	7.9415	−61.6495	−0.0220
Positive	303	6.0255	5.5143	0.0216	30.6083
8 to 14	**578**	**−5.8413**	**12.0551**	**−52.1272**	**36.4987**
Negative	395	−11.4749	9.8160	−52.1272	−0.0011
Positive	183	6.3186	5.8802	0.0273	36.4987
15 to 21	**573**	**−10.5470**	**13.6352**	**−66.8204**	**28.8170**
Negative	457	−14.7233	11.7841	−66.8204	−0.0236
Positive	116	5.9063	5.6330	0.0146	28.8170
More than 21	**658**	**−14.9267**	**17.2365**	**−94.6190**	**30.4210**
Negative	570	−18.1129	16.1738	−94.6190	−0.0722
Positive	88	5.7111	5.9503	0.0489	30.4210
Grand Total	**2382**	**−8.1896**	**14.6219**	**−94.6190**	**36.4987**

Source: Authors.

of mispricing is as high as ₹ −14.9267 and for 1–7 days, the mean is −0.4645. The under-pricing in futures also decreases with a reduction in DTE.

Our findings are in line with studies conducted earlier in Indian and foreign markets. Vipul (2005, 2008) observed that DTE have a significant impact on futures mispricing. Misra et al. (2006) also reported that arbitrage profits are more for far-month futures than for near-month futures. Yadav and Pope (1990), Bühler and Kempf (1995) and Fasaas (2010) reported similar results in foreign markets.

Vipul (2005) also offered a plausible explanation of this relationship. Accordingly, since the S&P Nifty futures are under-priced, there are many 'shorters' in the market. When expiry date approaches, they start winding up their positions, giving a boost to the liquidity in the market, which in turn reduces mispricing.

Mispricing with Respect to Basis

An attempt has been made to study the behaviour of mispricing in the conditions of negative basis (when spot price is less than futures price) and positive basis (when spot price is more than the futures price). From Table 5, it can be observed that out of 2,382 total observations, there were 1295 days (54.37 per cent) of negative basis and the remaining 1087 days of positive basis. It is evident that the mean magnitude of mispricing is much higher (₹ −17.933) for negative basis than for positive basis (₹ −0.0109). This indicates that the divergence between the fair price and observed price is more in terms of magnitude when the market displays the possibility of trending in future. But at the same time the mispricing is largely un-exploitable due to short-sales restrictions.

Table 5. Mispricing with Respect to Basis

	Observations	Mean	Standard Deviation	Min.	Max.
Spot Price < Futures Price	**1295**	**−0.0109**	**8.8529**	**−31.2204**	**36.4987**
NEGATIVE	605	−6.9155	6.4402	−31.2204	−0.0011
POSITIVE	690	6.0431	5.6803	0.0146	36.4987
Spot Price > Futures Price	**1087**	**−17.9333**	**14.1622**	**−94.6190**	**−0.2126**
NEGATIVE	1087	−17.9333	14.1622	−94.6190	−0.2126
Grand Total	**2382**	**−8.1896**	**14.6219**	**−94.6190**	**36.4987**

Source: Authors.

Table 6(a). Descriptive Statistics for Mispricing Data for the Period 2003–2012

N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
2382	−94.62	36.5	−8.1896	14.62185	−1.267	0.05	3.486	0.1

Source: Authors.

Table 6(b). One-Sample Kolmogorov–Smirnov Test for Normality Test for Mispricing Series

		Mispricing
N		2382
Normal Parameters[a,b]	Mean	−8.1896
	Std. Deviation	14.62185
Most Extreme Differences	Absolute	0.089
	Positive	0.058
	Negative	−0.089
Kolmogorov-Smirnov Z		4.362
Asymp. Sig. (two-tailed)		0.000

Source: Authors.
Notes: a. Test distribution is Normal.
b. Calculated from data.

Statistical Significance of Mispricing

From the above discussion it is evident that the observed price of CNX Nifty futures exhibit a departure from the fair price as determined by the cost-of-carry model. It remains to be seen whether these differences are statistically significant or not. We check the normality assumptions for the series of mispricing. Table 6(a) presents the descriptive statistics. The skewness and kurtosis indicate that the mispricing series may not be following the normal distribution. This is confirmed with a one-sample Kolomogrov–Smirnov (KS) test as reported in Table 6(b). The *p*-value is significant at the 5 per cent confidence level, thereby rejecting the null hypothesis that the distribution of mispricing is normal.

We propose to employ non-parametric tests due to non-normality of mispricing data. The Mann–Whitney U test for two samples and the Kruskal–Wallis (H statistics) test for k samples are used.

Kruskal-Wallis test confirms whether there a statistically significant difference between all samples exists, but it does not reveal the pair, which is different from the group. Hence a post-hoc study using Dunn's multiple comparison tests has been applied in case of testing the differences between various pairs of days to expiry. The outcomes are presented in Table 7(a). The results clearly indicate that there exists significant difference across various DTE with regard to mispricing. The post-hoc analysis, with Dunn's multiple comparisons (Table 7(b)), confirms that each pair is significantly different from the other at the 5 per cent significant level. This reiterates that mispricing differs significantly with days to expiry, within all groups. In terms of absolute values, too, Table 4 illustrated the same results.

Mann–Whitney U test was conducted to evaluate whether mispricing series is different under the situations

Table 7(a). Kruskal–Wallis Test for Differences in Mispricing across Groups of DTE for 2003–2012

K (observed value)	376.876
K (critical value)	7.815
DF	3
p-value (two-tailed)	<0.0001
alpha	0.05

Source: Authors.

Table 7(b). Dunn's Multiple Comparisons (Mean Rank Differences) across Groups of DTE for 2003–2012

DTE	Frequency	1 to 7	8 to 14	15 to 21	More than 21
1 to 7	573	0	329.277*	571.314*	720.330*
8 to 14	578	−329.277*	0	242.037*	391.053*
15 to 21	573	−571.314*	−242.037*	0	149.016*
More than 21	658	−720.330*	−391.053*	−149.016*	0

Source: Authors.
Note: *Denotes significance at the 5% level.

Table 8. Mann–Whitney test for Differences in Mispricing with Respect to Basis for 2003–2012

U	1266665.000
Expected value	703832.500
p-value (two-tailed)	<0.0001
alpha	0.05

Source: Authors.

of negative basis and positive basis. Looking at Table 8, it is clear that there exists a significant difference in mispricing with respect to the direction of basis. This finding is exclusive to our study in the context of the Indian market to the best of our knowledge. In operational terms, it implies that traders can analyse whether futures are at a discount or premium and devise arbitrage strategies accordingly. In the presence of short-sales restrictions, alternate strategies/instruments can be identified to make risk-free profits. From the viewpoint of the market, this denotes pricing inefficiency.

Conclusions

Conclusion

This article attempts to study the mispricing in the CNX Nifty index futures for the period 2003–2012. A total of 2382 observations were analysed for the two sub-periods so defined. Non-parametric tests were applied to test the significance of mispricing signals with respect to DTE and basis.

Our results indicate divergence between the observed futures price and fair price, as determined by the cost-of-carry model. The CNX Nifty index futures market was underpriced for the sample period at ₹ −8.1896. Although seemingly large in magnitude, they may not be profitable after considering transaction costs. The negative violations were more in terms of frequency and magnitude, indicating that a risk-free profit-making opportunity is not being arbitraged away, possibly due to short-sale restrictions.

Examining the trend for the study period, for the first sub-period (2003–2007), the mean under-pricing rose consistently. It seems reasonable to assume that since the derivatives market were at a nascent stage then; therefore it took time for the traders to understand the market mechanics and put in place arbitrage strategies. This was evident in the second sub-period (2008–2012). The index futures market improved in terms of pricing efficiency and by 2012 since the mean under-pricing stood at negligible ₹ −0.2281. The only point of alarm was in the year 2012 when the proportion of overpricing was more than under-pricing.

We tabulated the mispricing series across various groups of DTE. Statistically significant differences were recorded for all the pairs. This proves that the mispricing signals are diverse for different maturity groups. Accordingly, further the contracts are from DTE, more is the mispricing.

The pricing efficiency of the CNX Nifty index futures was also gauged in terms of the difference between spot price of the underlying index and its observed futures price. It was observed that under the situation of negative basis, the magnitude of mispricing was more than positive basis. Again the difference was found to be statistically significant.

Our findings can have important implications especially when the CNX Nifty index futures are being used as a hedging instrument. Also, when futures are used in conjunction with options or any other financial product for complex trading strategies, the under-pricing in the futures market may undercut potential benefits. Mispricing may also lead to poor price discovery in the market.

Limitations of the Study

These findings need to be interpreted with caution since we do not consider transaction cost or bid–ask spread in our analysis. Also, we have assumed that the cost-of-carry model is an efficient model of estimating futures prices and have adjudged the observed futures price accordingly. Any model cannot be without its fallacies, especially in terms of its assumptions. Our study utilises daily closing prices, and our results may suffer from the impact of non-synchronous trading.

Scope for Further Research

There are many possible extensions of this work. A similar study can be conducted for recent period for stock futures, currency futures and interest rate futures. The behaviour and cause and effect of mispricing can be studied with respect to volatility, open interest, volume, etc., for the past decade.

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